Use language identification dataset that you developed in the Assignment 1. Use the following classifiers and evaluate your model using macro-F1 evaluation metric. You cannot use any predefined classes under sklearn for the evaluation. You should use FastText wiki vectors for different languages. (Download it from here: https://fasttext.cc/docs/en/pretrained-vectors.html)

1. Logistic regression by averaging the word vectors and using them as features
2. Multi-Layer Perceptron with hidden layers n=0, 1, 2
3. LSTM
4. GRU
5. FastText Classification (https://fasttext.cc/docs/en/supervised-tutorial.html). Here you can use the FastText library.

Compare all classifiers and report the macro-F1 scores.